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Secure Communication Protocol for Unmanned Aerial Vehicles (UAVs)

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Introduction

1.1 Introduction

Unmanned Aerial Vehicles (UAVs) are widely used in civilian and military applications such as surveillance, disaster management, and logistics. Their operation depends on wireless communication between UAVs, Ground Control Stations (GCSs), and other system components. MAVLink is a lightweight protocol commonly used for UAV communication. However, wireless communication exposes UAV systems to threats such as spoofing, message injection, manipulation, and replay attacks. Therefore, this project proposes a modular security framework based on an independent Security Core that can be connected to UAV platforms such as PX4 through an adapter layer.

1.2 Problem Statement

UAV communication relies heavily on wireless networks, making it vulnerable to cybersecurity threats such as spoofing, message injection, message manipulation, and replay attacks. Although MAVLink provides lightweight and efficient communication, additional security mechanisms are required to protect UAV messages. Existing security approaches may be closely coupled to specific UAV platforms, limiting their flexibility and reuse. Therefore, this project addresses the need for a modular security framework that separates security mechanisms into an independent Security Core, allowing them to be developed, tested, and integrated with UAV platforms such as PX4 through an adapter layer.

1.3 Project Objectives

The main objective of this project is to design and implement a modular security framework for UAV communication that provides reusable security mechanisms independently of the underlying UAV platform.

The specific objectives of the project include:

• Analyzing the security vulnerabilities and threats affecting UAV communication systems.

• Studying existing security mechanisms used to protect UAV communication.

• Designing an independent Security Core for secure message processing.

• Implementing security mechanisms including authentication, message integrity, cryptographic protection, and anti-replay protection.

• Designing a modular interface that allows the Security Core to communicate with a UAV platform through an adapter layer.

• Integrating the security framework with a UAV communication environment, such as PX4, as a subsequent integration stage.

• Evaluating the security, performance, and effectiveness of the proposed framework.

1.4 Significance of the Study

The importance of this project lies in improving the security and reliability of UAV communication systems, which are increasingly used in civilian, industrial, and critical applications. Securing UAV communication helps protect transmitted information, reduce the risk of unauthorized message injection and manipulation, and improve the reliability of UAV operations. Furthermore, the proposed modular architecture separates security functionality from the underlying UAV communication platform. This separation can facilitate the testing, maintenance, reuse, and future extension of security mechanisms without requiring the entire communication system to be redesigned. The proposed framework also provides a foundation for future UAV security developments, including integration with PX4-based environments, secure UAV-to-Ground Station communication, and secure UAV-to-UAV communication. Therefore, the study contributes to the development of more flexible and secure communication frameworks for UAV systems.

Chapter 1: Theoretical Concepts

1.1 Introduction

Unmanned Aerial Vehicles (UAVs) have emerged as an important technology in a wide range of civilian and military applications, including surveillance, aerial photography, disaster management, environmental monitoring, and logistics. The increasing reliance on UAVs has consequently led to a growing dependence on wireless communication for the exchange of control commands, telemetry data, and operational information between UAVs, Ground Control Stations (GCSs), and other network components. Among the communication protocols widely adopted in UAV systems, MAVLink provides a lightweight and efficient mechanism for exchanging messages between system components. Its design is particularly suitable for applications that require low communication overhead and timely transmission of control and telemetry information. However, the wireless nature of UAV communication introduces security challenges, including eavesdropping, message injection, spoofing, message manipulation, and replay attacks. Consequently, securing UAV communication requires the integration of appropriate security mechanisms capable of providing authentication, message integrity, confidentiality, and protection against replay attacks while preserving the efficiency and reliability required by UAV communication systems.

1.2 UAV System Architecture

A UAV system consists of several components that cooperate to perform flight control, communication, monitoring, and mission-related operations. The main components generally include the UAV, flight controller, communication network, and Ground Control Station. The flight controller is responsible for processing control commands and sensor information and for performing the functions required to control the UAV. The Ground Control Station provides an interface for monitoring the UAV, sending commands, and managing missions. Communication between these components is commonly performed through wireless networks. Depending on the application, communication may occur between the UAV and the GCS or directly between multiple UAVs. Therefore, the communication architecture must consider both the operational requirements and the security risks associated with wireless communication. The wireless nature of UAV communication makes the system vulnerable to attacks such as eavesdropping, message injection, spoofing, manipulation, and replay attacks. Consequently, security mechanisms must be considered as an important part of UAV communication systems.

1.3 PX4 Autopilot Platform

PX4 is an open-source flight-control platform designed for autonomous and remotely operated vehicles. It uses MAVLink as a communication mechanism with Ground Control Stations and external components such as companion computers and MAVLink-enabled devices. This communication enables the exchange of telemetry, commands, status information, and other operational data between the flight controller and external systems [12], [14].

1.4 MAVLink Communication Protocol

MAVLink is a lightweight messaging protocol widely used for communication in UAV systems. It defines messages that can be exchanged between UAV components, autopilots, and ground stations. MAVLink supports communication over different underlying communication and transport technologies, which contributes to its flexibility and adoption in UAV applications [12]. MAVLink communication may carry telemetry information, vehicle status, navigation information, mission commands, and control-related messages. The protocol was designed with efficiency and low communication overhead in mind, which makes it suitable for UAV environments where communication bandwidth and computational resources may be limited [5], [12]. However, the security characteristics of MAVLink have been the subject of extensive research. Studies have identified vulnerabilities related to message confidentiality, authentication, message modification, replay, and command injection. Consequently, different security mechanisms have been proposed to strengthen MAVLink while maintaining its lightweight and real-time characteristics [1], [2], [5], [7], [8].

1.4.1 MAVLink Message Structure

Each transmitted packet follows a structured format inspired by the MAVLink protocol to ensure efficient parsing and communication reliability.



Figure1 : MAVLink Packet Structure [12]

Table 1: MAVLink packet Structure

Field	Size	Function
STX	1 byte	Start of frame (0xFD)
LEN	1 byte	Payload length
Incompat Flags	1 byte	Non-compatible feature flags
Compat Flags	1 byte	Compatible feature flags
Seq	1 byte	Sequence number
SysID	1 byte	System ID
CompID	1 byte	Component ID
MsgID	3 bytes	Message ID
Payload	Variable	Actual data
Checksum	2 bytes	Error detection (CRC)
Signature	13 bytes (optional)	Message authentication

1.4.2 HEARTBEAT Message

The HEARTBEAT message is one of the fundamental message types in the MAVLink communication protocol. It is periodically transmitted between UAV nodes and the Ground Control Station (GCS) to indicate node availability, communication status, and operational activity [13], [14]. The HEARTBEAT message enables communicating entities to detect active UAV systems, monitor connectivity status, and identify communication interruptions during UAV operation. In practical UAV communication environments, heartbeat messages are commonly utilized for system discovery, link monitoring, and real-time status synchronization between communicating nodes [13]. Since HEARTBEAT packets are transmitted periodically through wireless communication channels, they may become vulnerable to replay attacks, packet injection, or unauthorized manipulation in the absence of appropriate communication security mechanisms [1], [4].

Figure 2: illustrates the structure of the MAVLink HEARTBEAT message used for communication status monitoring between UAV nodes and the Ground Control Station. The message contains several operational fields including vehicle type, autopilot type, operating mode, system status, and MAVLink protocol version. These fields enable communication entities to monitor UAV activity, maintain connectivity awareness, and synchronize operational states during real-time UAV communication



Figure 2:MAVLink HEARTBEAT Message Structure [13]

Table 2MAVLink HEARTBEAT Message Fields

Field	Size	Description
Type	8 bits	Specifies the UAV platform type
Autopilot	8 bits	Indicates the autopilot system used by the UAV
Base Mode	8 bits	Represents the UAV operating mode flags
Custom Mode	32 bits	Defines custom UAV operational modes
System Status	8 bits	Indicates the current UAV operational status
MAVLink Version	8 bits	Specifies the MAVLink protocol version

1.4.3 MAVLink Communication with PX4

PX4-based UAV systems can use MAVLink to exchange information with a Ground Control Station and other system components. MAVLink messages may therefore form an important communication interface between the flight-control system and external applications [12]. From a security perspective, the communication interface can be protected by placing security mechanisms around or alongside the MAVLink communication process. Research has explored encryption and authentication mechanisms that protect MAVLink traffic while attempting to preserve its real-time communication characteristics [1], [5], [7], [9]. In the proposed framework, the security mechanisms are designed independently of PX4. A PX4 Adapter can subsequently provide the interface between the independent security layer and PX4/MAVLink communication. This separation supports portability and makes it possible to test the security mechanisms independently before platform integration.

1.5 Security Requirements in UAV Communication Systems

UAV communication systems require several security properties because communication may contain control commands, telemetry, navigation information, mission data, and other sensitive information. Research on UAV and MAVLink security commonly identifies confidentiality, authentication, integrity, availability, and protection against replay and related attacks as important requirements [2], [5], [8], [10], [11].

1.5.1 Entity Authentication

Entity authentication is the process of verifying the identity of a communicating entity. In UAV communication, authentication may be required between the UAV and the GCS or between two UAVs before sensitive information or commands are accepted. The need for mutual authentication becomes particularly important in UAV networks because an attacker may attempt to impersonate a legitimate UAV or GCS. Recent UAV security research has investigated lightweight mutual-authentication and key-agreement protocols designed to operate under the computational and communication constraints of UAV platforms [3], [10], [11]. In

a modular security framework, entity authentication is therefore treated as a distinct security service. Authentication establishes confidence in the identity of the communicating party, while message-level mechanisms subsequently protect individual messages.

1.5.2 Message Authentication

Message authentication verifies that a received message was generated by an authorized communicating entity and has not been modified by an attacker. It is different from entity authentication because authenticating an entity does not automatically guarantee that every subsequently received message is valid. Authenticated encryption and authentication mechanisms have been investigated for MAVLink to provide message authenticity while protecting against message injection and modification [7], [9]. Message authentication is therefore an important component of secure UAV communication, especially for command and mission-related messages.

1.5.3 Integrity

Integrity ensures that information has not been altered during transmission. In UAV systems, message modification may cause incorrect telemetry information to be processed or may alter commands transmitted from a GCS to a UAV. MAVLink security research identifies message modification and injection as significant threats. Cryptographic authentication mechanisms can provide integrity protection by allowing the receiver to detect unauthorized changes to protected information [5], [7], [8].

Integrity protection can apply to both payload information and security-relevant message metadata. AEAD mechanisms are particularly useful because associated data can be authenticated without necessarily being encrypted [7].

1.5.4 Confidentiality

Confidentiality prevents unauthorized parties from obtaining the content of transmitted information. Wireless UAV communication can expose telemetry, mission information, and control data to passive attackers if encryption is not used.

Several studies have investigated encryption as a means of protecting MAVLink communication. Different encryption algorithms have been evaluated according to their security and performance characteristics in UAV environments [1], [5], [7]. AEAD-based approaches are particularly relevant because they can combine confidentiality with authentication and integrity protection [7], [9].

1.5.5 Anti-Replay Protection

A replay attack occurs when an attacker captures a legitimate message and retransmits it later. In UAV systems, replayed commands or mission messages may cause an unwanted operation even if the original message was valid when it was first transmitted. Sequence numbers, timestamps, nonces, and other state information can be used to distinguish current messages from previously accepted messages. MAVLink security research identifies replay attacks as an important threat, while authentication protocols and secure UAV communication schemes incorporate mechanisms intended to prevent replayed communications [3], [8], [11], [12]. In the proposed framework, anti-replay protection is treated as an independent security requirement. Its detailed sequence-number and sliding-window.

1.5.6 Availability and Reliable Communication

Availability ensures that legitimate UAV communication remains usable when required. UAV systems may be affected by denial-of-service attacks, interference, packet loss, temporary disconnections, and other communication failures. Security mechanisms must therefore be considered together with communication reliability [5], [6], [10]. Reliable communication may require mechanisms that distinguish security rejection from ordinary packet loss. A secure system should reject malicious or invalid messages while still allowing legitimate communication to recover from transmission failures. This consideration motivates the use of reliable mission communication and controlled retransmission mechanisms in the proposed framework.

1.6 Cryptographic Mechanisms

Cryptography provides fundamental mechanisms for protecting information exchanged between communicating entities. In UAV communication systems, cryptographic mechanisms can be used to provide confidentiality, integrity, authentication, and secure key establishment. Research on secure MAVLink communication has investigated several classes of cryptographic algorithms, including symmetric encryption, asymmetric cryptography, authenticated encryption, and authentication mechanisms [1], [3], [7], [9],[11].

1.6.1 Symmetric Cryptography

Symmetric cryptography uses the same secret key, or related secret key material, for encryption and decryption. It is generally suitable for high-speed data protection because symmetric algorithms can be computationally efficient and require relatively low processing overhead. Several MAVLink security studies have evaluated symmetric encryption algorithms for protecting UAV communication. The evaluation of different algorithms is particularly important in UAV environments because computational resources, memory, latency, and energy consumption may be constrained [1], [5], [7]. ChaCha20 is one symmetric encryption mechanism investigated for UAV communication. Recent MAVLink research has also evaluated ChaCha20-Poly1305 as an authenticated-encryption mechanism and reported favorable execution-time characteristics in comparison with other evaluated AEAD schemes [7].

1.6.2 Asymmetric Cryptography

Asymmetric cryptography uses a pair of related keys, commonly referred to as a public key and a private key. It can be used for authentication, digital signatures, and secure establishment of shared cryptographic keys. Authentication and key-agreement protocols for UAV communication frequently use asymmetric cryptographic techniques to establish trust between UAVs and ground stations [3], [10], [11]. These mechanisms can reduce the need to distribute long-term symmetric encryption keys directly between every pair of communicating devices. However, asymmetric operations may introduce greater computational cost than

symmetric encryption. Therefore, UAV security protocols often attempt to combine asymmetric mechanisms for authentication or key establishment with efficient symmetric mechanisms for protecting the subsequent communication [3], [10].

1.6.3 Authenticated Encryption with Associated Data

Authenticated Encryption with Associated Data (AEAD) combines encryption and authentication in a single cryptographic construction. It provides confidentiality for encrypted data and integrity/authenticity protection for both encrypted data and associated data. AEAD is especially relevant to MAVLink because some message information may need to remain available for protocol processing while still being protected against unauthorized modification. Research specifically evaluating AEAD mechanisms for MAVLink demonstrated that AEAD can provide message confidentiality, integrity, and authenticity while also protecting relevant MAVLink header information through authenticated data [7]. The use of AEAD therefore provides a suitable theoretical basis for a security layer that must protect both message content and security-relevant metadata.

1.6.4 Digital Signatures

Digital signatures provide a mechanism for authenticating the origin of information and detecting unauthorized modification using asymmetric cryptography. A sender generates a signature using a private key, while a receiver verifies the signature using the corresponding public key. Digital signatures and certificate-based authentication are relevant to UAV networks where stronger identity assurance and key-management mechanisms may be required. Research on UAS networks discusses digital certificates, public-key infrastructure, authentication, and key management for secure UAV-to-UAV communication [6], [10].
However, the computational and communication overhead of public-key operations must be considered in resource-constrained UAV environments. Lightweight authentication research therefore investigates alternative mechanisms and optimized protocols for UAV systems [3], [10], [11].

1.7 ChaCha20-Poly1305

ChaCha20-Poly1305 is an AEAD construction that combines the ChaCha20 stream cipher with the Poly1305 message authentication algorithm. It provides confidentiality, integrity, and authentication within a single cryptographic operation [7].

1.7.1 ChaCha20 Encryption Algorithm

ChaCha20 is a modern symmetric stream cipher designed to provide high performance and strong security in software-based environments. It operates on a 512-bit internal state consisting of a 256-bit secret key, a 96-bit nonce, a 32-bit counter, and predefined constants. Through a sequence of quarter-round transformations involving modular addition, bitwise XOR, and rotation operations, ChaCha20 generates a pseudorandom keystream. Encryption is performed by applying a bitwise XOR operation between the plaintext and the generated keystream, resulting in ciphertext. Decryption follows the same process due to the symmetric nature of the algorithm. The security of ChaCha20 is based on the unpredictability of its keystream and resistance to known cryptanalytic attacks. Unlike traditional block ciphers that may require hardware acceleration for optimal performance, ChaCha20 is optimized for software execution, making it highly suitable for UAV systems with limited computational resources. Its consistent performance across different hardware platforms further enhances its applicability in real-time communication environments [1], [5], [7].

Equation (1): ChaCha20 Encryption is performed by applying a bitwise XOR operation between the plaintext message and the keystream generated using the ChaCha20 stream cipher. The generated keystream depends on the secret key, nonce value, and block counter in order to ensure secure and unique encryption for each transmitted packet.

$$C = P \oplus KS \quad (1)$$
$$KS = ChaCha20(Key, Nonce, Counter)$$

Where:

- C: Ciphertext

- P: Plaintext
- KS: Keystream generated by ChaCha20
- Key: secret encryption key
- Nonce: unique value used once
- Counter: block counter ensuring unique keystream blocks

Table 3: ChaCha20 Specifications

Parameter	Value	Description
Key Size	256 bits	Secret key
Nonce Size	96 bits (12 bytes)	Unique per packet
Block Size	512 bits	Internal structure
Operation	XOR	Encryption
Performance	High	Suitable for real-time

1.7.2 Poly1305 Authentication Algorithm

Poly1305 is a cryptographic message authentication code designed to ensure data integrity and authenticity. It operates by evaluating polynomial functions over a finite field defined by, producing a fixed-length authentication tag based on a secret key and the input message. Upon reception, the authentication tag is recomputed and compared with the transmitted value. Any mismatch indicates that the message has been altered or corrupted during transmission. When combined with ChaCha20, Poly1305 forms an authenticated encryption scheme known as ChaCha20-Poly1305, which belongs to the class of Authenticated Encryption with Associated Data (AEAD). This construction provides confidentiality, integrity, and authentication simultaneously and is widely adopted in modern secure communication systems due to its efficiency and strong cryptographic guarantees [7].

Equation (2): Poly1305 Authentication is used to generate a secure authentication tag for ensuring message integrity and sender authenticity. The authentication tag is generated using the secret key, ciphertext, and associated packet data before packet transmission.

$$TAG = Poly1305(K, C, A) \quad (2)$$

Where:

- TAG: Authentication tag
- K: Secret key
- C: Ciphertext
- A: Associated data (Header, SEQ)

Table 4: Poly1305 Specifications

Parameter	**Value**	**Description**
Key Size	256 bits	Authentication key
Tag Size	128 bits (16 bytes)	Integrity verification
Output	MAC Tag	Authentication
Security	High	Prevents tampering

The ChaCha20-Poly1305 authenticated encryption mechanism combines lightweight stream encryption with message authentication to provide confidentiality, integrity, and authenticity simultaneously. As illustrated in Figure 3, the encryption process begins by generating a pseudorandom keystream using the ChaCha20 algorithm based on a secret key and a unique nonce value. The generated keystream is then combined with the plaintext message using the XOR operation to produce the ciphertext. In parallel, a subkey derived from the ChaCha20 keystream is utilized by the Poly1305 authentication algorithm to generate a 128-bit authentication tag. This authentication tag ensures message integrity and enables the receiver to detect any unauthorized modification or packet tampering during transmission. The integration of ChaCha20 and Poly1305 forms an efficient Authenticated Encryption with Associated Data (AEAD) scheme widely adopted in modern secure communication systems due to its strong security guarantees and low computational overhead.

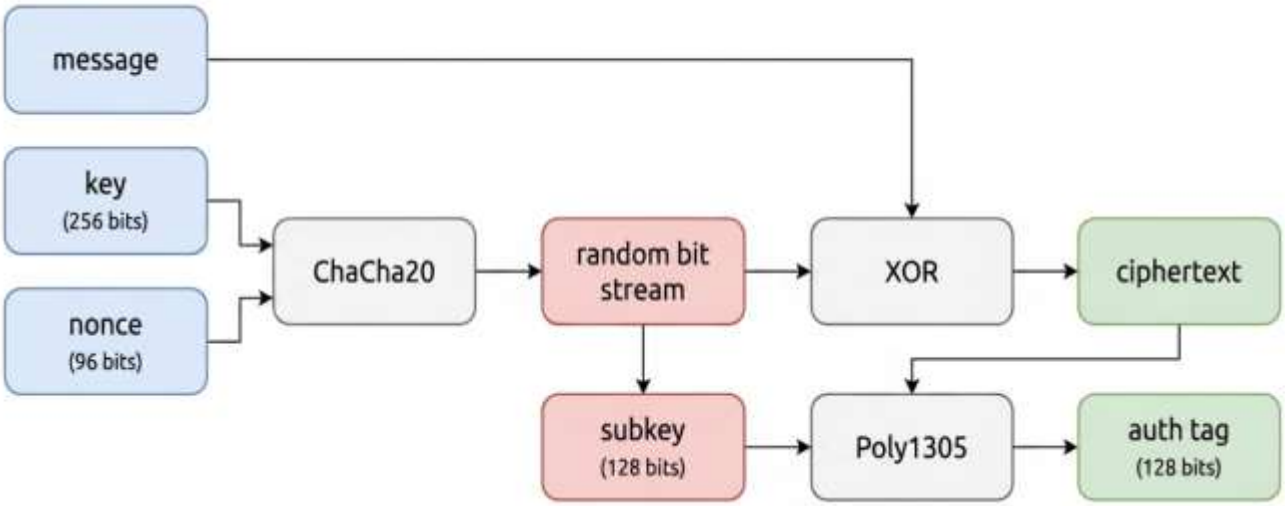


Figure3:ChaCha20-Poly1305 Encryption [4]

Equation (3): AEAD Models generates encrypted ciphertext together with an authentication tag to provide confidentiality and integrity protection for MAVLink packet transmission.

$$(C, TAG) = AEAD_ChaCha20_Poly1305(K, Nonce, P, A) \quad (3)$$

The nonce (12 bytes) and authentication tag (16 bytes) are directly implemented using OpenSSL functions.

1.7.3 Associated Data

Associated Data refers to information that is authenticated but not encrypted. This mechanism is useful when certain message fields must remain available for routing, processing, or protocol identification while still requiring integrity protection. In MAVLink security research, AEAD has been investigated as a means of protecting the MAVLink payload while also providing integrity protection for relevant header information through associated data [7]. This concept provides a theoretical basis for protecting security-relevant message

metadata without necessarily encrypting every field. In UAV communication research, ChaCha20 has been evaluated as an encryption mechanism for MAVLink traffic. Studies have compared its performance with other encryption approaches under UAV or simulated UAV environments, considering factors such as execution time, CPU utilization, memory consumption, and communication performance [1], [5], [7].

1.8 Authentication and Key Management Concepts

Authentication and key management are fundamental components of secure UAV communication. Authentication establishes trust between communicating parties, while key management determines how cryptographic keys are generated, distributed, established, stored, refreshed, and potentially revoked. Recent UAV security research emphasizes the importance of secure key establishment and lightweight authentication because UAV networks may involve multiple devices, dynamic topology, limited resources, and wireless communication links exposed to attackers [3], [10], [11].

1.8.1 Entity Authentication

Entity authentication verifies that a communicating UAV, GCS, or other network participant is the entity it claims to be. In UAV networks, mutual authentication may be required so that both sides can verify each other's identity before establishing protected communication. Recent authentication protocols for UAV communication have proposed lightweight mutual-authentication and key-agreement mechanisms designed to reduce computational cost while maintaining security properties [3], [10], [11].

1.8.2Message Authentication

Message authentication focuses on individual messages rather than only on the identity of the communicating entity. A message-authentication mechanism allows the receiver to determine whether the message was generated by an authorized source and whether it was modified during transmission. Authenticated encryption provides this property together with confidentiality, while digital signatures and other authentication mechanisms can provide message authenticity through

asymmetric cryptography [6], [7], [9].

1.8.3 Key Management

Key management concerns the complete lifecycle of cryptographic keys. This includes key generation, distribution or agreement, storage, use, renewal, and revocation. Key management is particularly challenging in UAV networks because multiple UAVs may communicate dynamically with GCSs and with other UAVs. Research has therefore investigated authenticated key-agreement protocols, certificate-based mechanisms, PKI, and lightweight approaches suitable for UAV environments [3], [6], [10], [11]. A distinction can also be made between long-term identity keys and temporary session keys. Long-term credentials are generally used to establish and maintain the identity of communicating entities, while session keys may be established to provide cryptographic protection for a particular communication session.

1.9 Anti-Replay Protection

Replay protection is required to prevent an attacker from successfully reusing a previously valid message. This requirement is particularly important for UAV control and mission communication because a previously valid command may become unsafe or inappropriate when executed at a later time. MAVLink security research identifies replay attacks as a relevant threat, while recent authentication and secure communication protocols incorporate mechanisms intended to detect and reject replayed messages [3], [8], [11], [12].

1.9.1 Sequence Numbers

A sequence number is a value associated with a message to indicate its position in a communication sequence. A receiver can use sequence information to determine whether a received message is new, duplicated, reordered, or older than previously accepted communication. Sequence-based protection is useful for real-time UAV communication because it allows the receiver to maintain communication state without requiring every message to contain a large amount of additional information [12].

1.9.2 Sliding-Window Mechanism

A sliding-window mechanism allows a receiver to accept a range of sequence numbers rather than requiring messages to arrive in exactly increasing order. This is useful in wireless networks where packets may be delayed or reordered.
The receiver maintains information about the highest accepted sequence number and the range of recently accepted sequence values. Messages outside the acceptable window can be rejected as stale, while duplicate sequence values can also be detected [13].

1.9.3 Duplicate, Reordered, and Out-of-Window Messages

Wireless communication does not always guarantee ordered packet delivery. Consequently, a secure receiver must distinguish between legitimate reordered messages and malicious replayed or stale messages. A duplicate message has already been accepted, while an out-of-window message is sufficiently old that it should no longer be accepted. A reordered message may still be legitimate if it falls within the permitted reception window. These distinctions are important for maintaining both security and communication reliability [13].

1.10 Reliable Mission Communication

Security alone does not guarantee reliable mission execution. UAV communication can experience packet loss, temporary disconnections, interference, or transmission failures. Therefore, mission-level communication may require mechanisms for determining whether an operation has been completed and for recovering from lost communication. UAV communication research emphasizes the importance of reliable and low-latency communication, particularly for safety-critical and cooperative UAV operations [6], [10]. Security mechanisms must therefore operate without unnecessarily preventing legitimate recovery from ordinary communication failures.

1.10.1 Mission Completion

Mission completion provides an application-level indication that a requested operation has been successfully executed. Instead of requiring an acknowledgment

for every individual communication packet, a mission-oriented design can use a completion message to indicate that the requested operation has reached its expected state. For a secure system, the mission-completion message must itself be authenticated and protected. Otherwise, an attacker could forge a completion message and cause the sender to incorrectly assume that a mission has been completed [12].

1.10.2 Timeout and Retransmission

When a required completion message is not received within an expected period, the sender may use a timeout mechanism to determine that communication may have failed. A retransmission can then be initiated according to the application's reliability requirements. Retransmission must be distinguished from security verification. A retransmission mechanism attempts to recover from communication loss, whereas anti-replay protection determines whether a received message is new and acceptable. Therefore, retransmission should not bypass authentication, integrity verification, or anti-replay checks [14].

1.11 UAV-to-UAV Secure Communication

Modern UAV systems increasingly support cooperative operations in which multiple UAVs exchange information directly or through network infrastructure. UAV-to-UAV communication may be required for collision avoidance, formation control, distributed sensing, mission coordination, and other cooperative applications. Security requirements for UAV-to-UAV communication are similar to those of UAV-to-GCS communication but may be more challenging because the communicating entities can change dynamically. Each UAV may need to authenticate another UAV, establish appropriate cryptographic protection, and verify the integrity and freshness of received messages. Research on UAS networks has investigated authentication, key management, authorization, zero-trust concepts, and secure V2V communication. These approaches demonstrate the importance of treating UAV-to-UAV links as security-sensitive communication channels rather than relying exclusively on the security of a GCS connection. The proposed framework therefore considers UAV-to-UAV communication as a natural extension of the security architecture. The detailed communication scenario, message

exchange, authentication process, and implementation are presented in the practical design chapter [6], [10], [11].

1.12Chapter Summary

This chapter presented the theoretical concepts required for understanding secure UAV communication. It introduced the UAV system architecture, PX4 as a UAV autopilot platform, and MAVLink as a lightweight communication protocol used in UAV systems. The chapter also discussed the security requirements of UAV communication, including entity authentication, message authentication, integrity, confidentiality, anti-replay protection, availability, and reliable communication The chapter further reviewed cryptographic mechanisms, with particular emphasis on symmetric cryptography, asymmetric cryptography, AEAD, digital signatures, and the ChaCha20-Poly1305 construction. Authentication and key-management concepts were also discussed, followed by sequence numbers and sliding-window mechanisms for replay protection. Finally, reliable mission communication and UAV-to-UAV secure communication were introduced as important components of a complete UAV security architecture. These theoretical concepts provide the foundation for the security framework design and implementation presented in the following chapters.

Chapter 2: Literature Review

2.1 Related Work

Recent years have witnessed significant research efforts aimed at enhancing the security of UAV communication systems, particularly those relying on the MAVLink protocol. Due to the lightweight nature and widespread adoption of MAVLink in platforms such as PX4 and ArduPilot, researchers have increasingly focused on developing secure communication frameworks capable of protecting telemetry and control messages against cyberattacks including replay attacks, spoofing, packet injection, and denial-of-service attacks. Khan et al. proposed an additional security layer for MAVLink communication to improve the confidentiality and integrity of telemetry data exchanged between UAVs and Ground Control Stations (GCSs). Their framework introduced encryption mechanisms directly into MAVLink packet transmission while preserving low communication overhead. Dixit et al. presented MAVShield, a lightweight cryptographic framework specifically designed for securing MAVLink-based UAV communication. The authors evaluated multiple encryption algorithms including AES-CTR, ChaCha20, Speck-CTR, and Rabbit cipher using a real UAV testbed environment. Gómez Rey and Hernández Troncoso investigated the performance of AEAD-based encryption algorithms for securing MAVLink communication. Their study compared AES-GCM, ChaCha20-Poly1305, and TLS 1.3 using a Parrot Bebop UAV platform. He et al. proposed a lightweight authentication and key-agreement protocol for UAV communication systems focusing on minimizing computational complexitywhile resisting replay and impersonation attacks. Aljumah introduced a multilevel authentication framework for secure UAV communication within Internet of UAVs (IoUAV) environments using layered authentication and lightweight cryptographic operations. Although these studies achieved considerable progress in UAV communication security, most existing approaches primarily focus on encryption and authentication while providing limited protection against replay attacks and duplicated packet execution.

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Weaknesses	Relevance to our Project
Security Strategies for Unmanned Aircraft Systems Networks	2023	IEEE ICC Workshop	Marco Hernandez; Gurkan Gur; Kamesh Namuduri	Proposed a comprehensive security framework for UAS networks and analyzed threats such as spoofing, jamming, and DoS using a Zero-Trust approach.	Threat classification; vulnerability analysis of UAV communication layers	Comprehensive overview of UAV security challenges	Focuses on threat analysis rather than practical implementation of a complete security .framework	Helps define the threat model for securing UAV–GCS communication
Security-Enhanced Lightweight Authentication Key-Agreement Protocol for UAV Communication	2025	Applied Sciences (MDPI)	Zhoucan He et al.	Introduced a lightweight authentication and key-agreement protocol suitable for resource-constrained UAV systems.	Computation cost; authentication delay; ECC operation count	Low computational overhead and resistance to replay/impersonation attacks	Focuses on authentication and key agreement rather than comprehensive message protection.	Supports designing lightweight authentication for UAV communication
Enhancing MAVLink Security in UAV Systems	2025	University Thesis / Research Study	Daniel Gomez Rey; Juan Pablo Hernandez Troncoso	Implemented encryption for MAVLink communication using AES-GCM, ChaCha20-Poly1305 and TLS 1.3.	Latency; RAM usage; communication overhead; battery consumption	Real UAV implementation and comparison of encryption methods	Focuses on encryption methods; comprehensive anti-replay and retransmission integration is not established.	Directly relevant for securing MAVLink-based UAV communication
A Secure Communication Protocol for Unmanned Aerial Vehicle Networks	2022	Conference / Research Paper	Various Authors	Designed a secure protocol with encryption and mutual authentication to protect UAV network communication.	Authentication time; key generation cost; communication overhead	Improves confidentiality and resistance to network attacks	Integration with MAVLink 2 and PX4 is not established in the available summary.	Provides protocol-level design ideas for secure UAV communication
UAV-Based Secure Data Communication: Multilevel Authentication Perspective	2024	Sensors (MDPI)	Abdullah Aljumah	Proposed a multi-level authentication model for Internet of UAVs using hashing, encryption and XOR operations.	Latency; throughput; packet loss; energy consumption	Layered authentication with improved efficiency	Emphasizes authentication without establishing a complete AEAD-based Security Envelope.	Useful for designing multi-layer authentication mechanisms

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Weaknesses	Relevance to our Project
A Novel Cipher for Enhancing MAVLink Security	2024	Journal / Conference in UAV Cybersecurity	Various Authors	Proposed a new encryption algorithm to protect MAVLink messages and prevent data interception or modification.	Encryption and decryption time; algorithm efficiency	An algorithm tailored to the MAVLink protocol	Focuses on encryption; comprehensive authentication and anti-replay mechanisms are not established.	Useful for developing a lightweight encryption layer for the project
MAVLink Protocol: A Survey of Security Threats and Countermeasures	2024	IEEE ICoDT2	Various Authors	Presented a comprehensive survey of MAVLink security threats and countermeasures, covering MAVLink 1/2, confidentiality, integrity, availability, authentication, encryption and key management.	Threat classification; security mechanisms; protocol and key-management analysis	Broad and systematic coverage of MAVLink security	Reviews existing solutions but does not provide a practical implementation of the proposed Security Core.	Provides the theoretical and State-of-the-Art foundation for the project's security architecture
Provably Secure Authentication Protocol for Traffic Exchanges in UAVs	2024	Research Paper	Various Authors	Proposed a lightweight authentication protocol providing mutual authentication, session-key agreement, anonymity and unlinkability, with formal security analysis.	Computation cost; communication overhead; security/provability analysis	Addresses replay, impersonation, MITM and other attacks while supporting session-key security properties	Focuses on authentication and key agreement rather than complete message protection.	Useful for the Entity Authentication, Key Management and Anti-Replay components
A Lightweight Authentication Protocol for UAVs Based on ECC Scheme	2023	Drones (MDPI), 7, 315	Zhang; Liu; Han; Yang	Proposed LAPEC, an ECC-based identity authentication and key-agreement protocol for UAVs, including backward secrecy of session keys.	Authentication and key-agreement computation; communication overhead; security analysis	Uses ECC to balance security and resource efficiency and improves session-key security	Focuses on ECC-based authentication and key agreement rather than integrated AEAD protection.	Supports asymmetric authentication and future session-key establishment in the Security Core

Article Title	Publication Year	Journal / Source	Authors	What Was Achieved in This Paper	Reported Performance Metrics	Strengths	Weaknesses	Relevance to our Project
SLAKA-IoD: A Secure and Lightweight Authentication and Key Agreement Protocol for Internet of Drones	2024	Drones (MDPI), 8, 374	Yuelei Xiao; Yu Tao	Proposed SLAKA-IoD, a lightweight authentication and key-agreement protocol using PUF, XOR, and hash functions. It supports mutual authentication and secure session-key establishment between drone–GS and drone–drone pairs, with informal and formal security analysis and Scyther verification.	Computation and communication cost; security-property analysis; comparison with related IoD authentication/key-agreement schemes	Supports D2G and D2D mutual authentication, session keys, replay resistance, and lightweight cryptographic operations for resource-constrained drones.	Focuses on PUF-based authentication and key agreement rather than a complete AEAD-based security framework.	Highly relevant to the Security Core Authentication and Key Management components and to the planned D2D secure-mission scenario. It supports moving beyond a static shared key toward session-key establishment.
Lightweight Drone-to-Ground Station and Drone-to-Drone Authentication Scheme for Internet of Drones	2025	Symmetry (MDPI), 17, 556	Qi Xie; Jilin Zhang	Proposed a lightweight D2G and D2D authentication and key-agreement protocol using PUFs. It enables direct mutual authentication between drones without GS involvement and addresses capture, impersonation, privacy, and forward-secrecy concerns.	Average computational-efficiency improvement of 39.44% versus comparable schemes; storage overhead during GS authentication; formal security analysis	Direct D2D authentication without GS, PUF-based protection against capture attacks, perfect forward secrecy, and improved efficiency.	Focuses on authentication and key agreement rather than a complete, modular Security Core.	Extremely relevant to the planned D1→D2 secure communication and mission-completion workflow, and to the future Authentication/Key-Management layer of the Security Core.

2.2 Research Gap

Despite the extensive research conducted on securing UAV communication systems, several important limitations remain insufficiently addressed. Most existing approaches focus mainly on encryption mechanisms to provide confidentiality and integrity for MAVLink packets. However, encryption alone cannot prevent replay attacks in which an attacker captures valid packets and retransmits them at a later time to trigger unauthorized command execution. Many previous studies also rely heavily on simulation-based evaluations or theoretical analyses without implementing practical replay attack scenarios involving packet capture and retransmission. Another significant limitation concerns performance evaluation under lightweight UAV communication constraints. Some proposed solutions introduce noticeable computational, bandwidth, and latency overhead that may negatively affect real-time telemetry communication and flight stability. Furthermore, limited research has explored modular secure communication architectures that can later be integrated into real UAV ecosystems such as PX4 autopilot systems and telemetry RF communication environments.

2.3 Research Contributions

This research proposes a lightweight secure communication framework for MAVLink-based UAV systems. The primary contributions of this work are summarized as follows:

A. Designing a secure MAVLink communication architecture using the ChaCha20-Poly1305 authenticated encryption algorithm.

B. Implementing an anti-replay protection mechanism based on sequence numbers and sliding-window validation.

C. Simulating a replay attack scenario using an attacker node capable of retransmitting captured encrypted MAVLink packets.

D. Measuring network performance metrics including latency, throughput, packet loss, and acknowledgment success rate.

E. Developing a visual UAV communication environment using NetAnim and Wireshark integration.

F. Designing the framework with modularity and extensibility to support future PX4 and telemetry RF integration.

2.4 Recent Advances in MAVLink and UAV Communication Security

Recent advances in UAV security research have increasingly focused on lightweight authenticated encryption algorithms optimized for embedded and resource-constrained systems. AEAD algorithms such as AES-GCM and ChaCha20-Poly1305 have become highly attractive due to their ability to simultaneously provide encryption and authentication with relatively low computational overhead. ChaCha20-Poly1305 has received particular attention because of its efficiency in software-based implementations where hardware acceleration may not be available. Another active research direction involves lightweight authentication and key-agreement protocols capable of securely establishing session keys under strict timing and computational constraints. Researchers have also explored zero-trust architectures, secure telemetry routing, V2V authentication, and secure communication standards for Beyond Visual Line of Sight (BVLOS) UAV operations.

2.5 Challenges

Securing UAV communication systems presents several technical and operational challenges. UAV platforms are inherently resource constrained in terms of CPU capability, memory capacity, and battery power. UAV telemetry and control communication require extremely low latency and high reliability because flight commands and telemetry data are time-sensitive. MAVLink communication commonly operates over UDP or serial telemetry links, both of which lack built-in reliability and security guarantees. Replay attacks remain another major challenge because encryption alone cannot prevent attackers from retransmitting previously

captured valid packets. Finally, real-world telemetry RF communication introduces practical constraints including limited bandwidth, high packet loss rates, interference, and communication instability.

2.6 Practical Applications

Secure MAVLink communication frameworks can be applied across numerous UAV domains requiring protected telemetry and command exchange. These include military reconnaissance, border surveillance, disaster response, smart agriculture, environmental monitoring, infrastructure inspection, emergency rescue operations, aerial mapping, and intelligent transportation systems.The proposed secure communication architecture helps protect UAV control commands and telemetry data against eavesdropping, spoofing, replay attacks, unauthorized command injection, and packet manipulation attacks. Furthermore, the modular design of the proposed framework enables future integration with PX4 autopilot systems, telemetry RF communication modules, and real UAV hardware platforms.

Chapter 3: Design and Implementation of a Secure MAVLink Communication System for UAVs

3.1 Introduction

This chapter presents the design and implementation methodology of the proposed security framework for UAV communication. The framework introduces an independent Security Core that provides cryptographic protection, authentication, integrity verification, and anti-replay mechanisms. ChaCha20-Poly1305 is adopted for authenticated encryption, while a dedicated PX4 Adapter is planned for subsequent integration with MAVLink 2 and PX4. The chapter also outlines the implementation stages and the methodology for evaluating security and performance.

3.2 Proposed Security Framework

The proposed framework introduces a modular Security Core to protect communications between Unmanned Aerial Vehicles (UAVs) and Ground Control Stations (GCSs), as well as direct UAV-to-UAV communications. The framework is designed to operate independently of PX4, allowing the Security Core to be developed and tested separately before integration through a dedicated PX4 Adapter. It consists of several interconnected layers responsible for application operations, security processing, message handling, and data transmission.

❖ Application Layer

The Application Layer manages UAV operations and MAVLink-based communication. It includes mission management, command processing, and telemetry handling. This layer generates outgoing messages and processes incoming messages, relying on the Security Core to provide the required protection during transmission.

❖ Security Core

The Security Core is the central component responsible for securing messages. It incorporates cryptographic protection, authentication, integrity verification, and anti-replay mechanisms. Its main components include:

- Key Management: Manages the cryptographic keys required for secure communication.
- Crypto Engine: Uses ChaCha20-Poly1305 to provide confidentiality, integrity, and cryptographic message authentication.
- Nonce Management: Ensures that nonces are managed correctly for secure encryption.

- Authentication Tag Verification: Verifies the authentication tag to detect message modification and reject invalid messages.
- Anti-Replay Protection: Uses sequence numbers and a sliding window to detect duplicate or outdated messages.

Entity authentication is handled through a separate mechanism to verify the identities of communicating UAVs and ground stations.

❖ Message Processing Layer

The Message Processing Layer manages the construction, encoding, decoding, and validation of messages. It includes message building, serialization, deserialization, header validation, and parsing. These functions ensure that messages are correctly structured and processed before being transmitted or delivered to the application.

❖ Transport Abstraction Layer

The Transport Abstraction Layer separates the Security Core from the underlying communication mechanism. This separation allows different transport technologies to be supported without requiring changes to the core security functions, thereby improving modularity and facilitating future extensions.

❖ Network / Transport Layer

The Network / Transport Layer carries messages between UAVs and ground stations through the available communication links. The security framework is designed to provide protection independently of the underlying transport technology.

❖ Supporting Modules

Supporting modules provide services that improve the framework's reliability and maintainability. These include configuration management, logging and monitoring, error handling, and memory safety. They support the operation of the framework without replacing its core security mechanisms.

❖ Communication Process

Consider a scenario in which UAV D1 sends a message to UAV D2. The Application Layer generates the message, and the Message Processing Layer

prepares it for transmission. The Security Core then applies the required cryptographic protection using ChaCha20-Poly1305. The protected message is transmitted through the transport and network layers to the receiving UAV.

Upon reception, the message is parsed and validated, and its cryptographic authentication tag is verified. The anti-replay mechanism then checks whether the message is acceptable based on its sequence number and the receiver's current replay window. Only messages that pass the required security checks are delivered to the application for processing. Invalid, modified, or replayed messages are rejected.

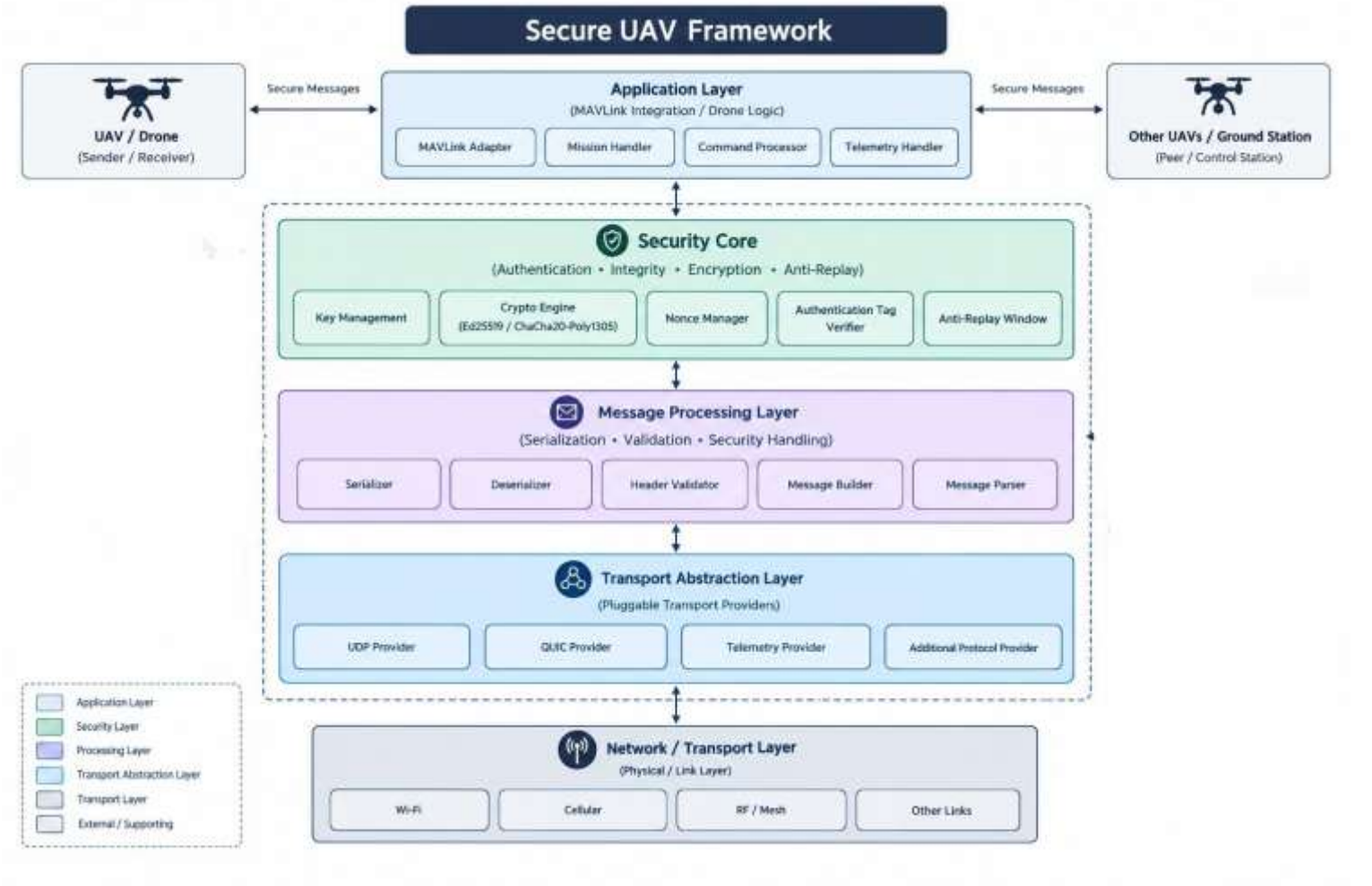


Figure3 :proposed framework

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